

A system solution

A solution of a system must satisfy every equation simultaneously.

$$x + y = 7 \quad x - y = 1$$

Does (4, 3) satisfy the system?

Checking both equations

$$2x + y = 8 \quad x - 2y = 1$$

Is (3, 2) a solution of the entire system?

Intersections of graphs

$$y = x + 1 \quad y = -x + 5$$

What ordered pair belongs to both lines?

Graphical solution

$$y = 2x - 1 \quad y = -x + 5$$

What intersection appears when you sketch both lines?

One solution

$$y = 3x + 2 \quad y = -2x + 7$$

What graph feature guarantees exactly one intersection?

Substitution

$$y = 4x - 3 \quad x + y = 12$$

What is the solution using substitution?

Substitution with a coefficient

$$x = 2y + 1 \quad 3x - y = 13$$

What is the solution using substitution?

Choosing a variable

$$2x + y = 9 \quad 5x - 3y = 6$$

Which equation makes substitution simplest, and what solution results?

Addition method

$$3x + y = 11 \quad 2x - y = 4$$

What is the solution after eliminating y ?

Scaling one equation

$$2x + 3y = 13 \quad x - y = 4$$

What is the solution using elimination?

Scaling both equations

$$3x + 2y = 12 \quad 2x - 3y = -5$$

What is the solution using elimination?

Fractional coefficients

$$\frac{x}{2} + y = 5 \quad x - y = 1$$

What is the exact ordered-pair solution?

An inconsistent system

An inconsistent system has no solution.

$$y = 2x + 1 \quad y = 2x - 4$$

Why does this system have no solution?

A dependent system

A dependent linear system has equations describing the same line.

$$x + 2y = 6 \quad 2x + 4y = 12$$

What entire set of ordered pairs solves this system?

A contradiction after elimination

$$2x - y = 3 \quad 4x - 2y = 9$$

What does elimination reveal about the solution set?

An identity after elimination

$$3x + y = 7 \quad 6x + 2y = 14$$

How can the infinite solution set be written using x as a parameter?

Counting linear intersections

Consider the solution set of a system of two linear equations in two variables.

Can two lines in the plane have exactly two distinct common points?

Selecting a method

$$5x - y = 8 \quad 2x + y = 13$$

Which method seems efficient, and what is the solution?

A ticket system

A show sells 80 tickets for \$620. Adult tickets cost \$10 and student tickets \$6. Every ticket is one of these types.

What system represents the numbers of adult and student tickets?

A mixture system

Mix a 20% salt solution and a 50% salt solution to obtain 12 liters of 30% solution. Assume volumes add.

How many liters of each solution are needed?

A break-even point

C and R are dollars; q is the number of items produced and sold, a nonnegative integer.

$$C(q) = 240 + 6q \quad R(q) = 10q$$

At what quantity does revenue equal cost?

Connecting equations to functions

$$f(x) = x + 4 \quad g(x) = 3x - 2$$

How does solving $f(x) = g(x)$ identify the solution of $y = f(x)$, $y = g(x)$?